

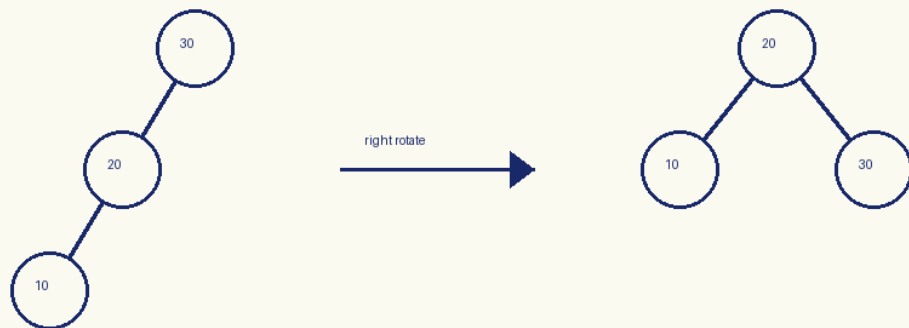
Algorithm Analysis: Tree Balances

This section documents self-balancing binary search trees. Notice the handwritten algorithm and rotation diagram below.

AVL Rotations

```
struct Node * rotRight(struct Node * y){  
  
    struct Node * x = y->left  
  
    y->left = x->right  
  
    x->right = y;  
  
    return x;  
}
```

LL case (left-left heavy):



$bf(30) = h(L) - h(R) = 2 \rightarrow$ unbalanced

after rotation all $bf = 0$, balanced